CS231n Lecture 2, accessible version (slides 22, 23 and 52)

Generated by the prototype pipeline from the original untagged PDF.

Slide 22: First classifier: Nearest Neighbor

Overview

A diagram explaining the nearest neighbour classifier. Five labelled training photos sit on the left, one unlabelled query photo of a cat on the right, and below them a distance metric that compares two images.

Parts

- Training data with labels: five small photos labelled deer, bird, plane, cat and car.
- Query data: a larger photo of a grey tabby cat, marked with a question mark because its label is unknown.
- Distance metric: the query cat and the training cat are placed between vertical bars, and an arrow maps the pair to the real numbers.

How the parts relate

The classifier compares the query image with every training image using the distance metric. The training image with the smallest distance gives its label to the query, so here the query would be labelled cat.

Formulas

The distance metric takes two images and returns one real number.

$$\| I_{\text{query}}, I_{\text{train}} \| \rightarrow \mathbb{R}$$

Slide 23: Distance Metric to compare images

Overview

A worked example of the L1 distance between two images. A formula at the top, then three 4 by 4 grids of pixel values: test image minus training image equals the pixel-wise absolute differences, which add up to 456.

Parts

- The L1 distance formula.
- Grid 1, test image pixel values.
- Grid 2, training image pixel values.
- Grid 3, the absolute difference of each pair of pixels.
- An arrow labelled add pointing to the total, 456.

How the parts relate

Each cell of grid 3 is the absolute difference of the matching cells in grids 1 and 2. For example the top left cell is 56 minus 10, which is 46. Adding all 16 differences gives the L1 distance of 456.

Formulas

L1 distance formula.

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

Tables

Test image pixel values

	Column 1	Column 2	Column 3	Column 4
Row 1	56	32	10	18
Row 2	90	23	128	133
Row 3	24	26	178	200
Row 4	2	0	255	220

Training image pixel values

	Column 1	Column 2	Column 3	Column 4
Row 1	10	20	24	17
Row 2	8	10	89	100
Row 3	12	16	178	170
Row 4	4	32	233	112

Pixel-wise absolute value differences (these add up to 456)

	Column 1	Column 2	Column 3	Column 4
Row 1	46	12	14	1
Row 2	82	13	39	33
Row 3	12	10	0	30
Row 4	2	32	22	108

Slide 52: Parametric Approach: Linear Classifier

Overview

A diagram of a linear classifier. An image of a cat goes into a function f of x and W, which outputs 10 class scores. Above it, the equation f of x W equals W x plus b, with the size of every term.

Parts

- Input: a photo of a cat, an array of 32 by 32 by 3 numbers, 3072 numbers in total.
- The function f of x and W, with x shown in blue and W in red.
- W, the parameters or weights, feeding into f from below.
- Output: 10 numbers giving class scores.

How the parts relate

The image x is flattened to a 3072 by 1 column. Multiplying by the 10 by 3072 weight matrix W and adding the 10 by 1 bias b gives a 10 by 1 vector, one score for each class.

Formulas

Linear classifier equation.

$$f(x, W) = Wx + b$$

The same equation with the size of every term.

$$\underbrace{f(x, W)}_{10 \times 1} = \underbrace{W}_{10 \times 3072} \underbrace{x}_{3072 \times 1} + \underbrace{b}_{10 \times 1}$$